Calibration Uncertainty in Adaptive Large Language Model Evaluation: Theory and Empirical Evidence

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Abstract

Adaptive testing with item response theory requires item parameters calibrated on other systems, but new, private or expensive language-model benchmarks often have only a handful of them. We study this calibration uncertainty with EOST-M (Entropy-Oracled Information-Optimal Self-Testing, Marginalized), which uses each item's expected response function under an empirical-Bayes prior and selects items by expected information gain. This model is exact given the calibrators' abilities and hyperparameters, and Gaussian uncertainty s in an item's difficulty caps its Fisher information at $2/(\pi s^2)$ for any response link. Two frozen-plan studies on eight benchmarks give three messages. First, plug-in Fisher-information testing fails with small panels: with eight calibrators and one target per family cluster, EOST-M predicted held-out verdicts better (median 5.9%, significant on 5 of 8 benchmarks). Second, accounting for the uncertainty removes much of this failure: EOST-M also beat a regularized Rasch-type test (2.4%), Bayesian tests that also integrate the calibrators' abilities were not significantly different from it, and its score error was lower than with random items under the same model, ATLAS-style testing, tinyBenchmarks and MINCE on every benchmark, meeting the frozen criterion only against MINCE. Third, the information criterion then matters little. EOST-M's 90% intervals still under-cover (0.74; 0.86 after recalibration).

Keywords: adaptive testing, item response theory, large language model evaluation, calibration uncertainty, expected information gain

1 Introduction

Large language models (LLMs) are compared by running them on static benchmarks and reporting an average. This evaluation is computationally expensive and statistically inefficient, because individual items differ substantially in the information they provide about a system's capabilities (Lalor et al., 2016; Rodriguez et al., 2021). Computerized adaptive testing (CAT), the psychometric answer to the same problem for human examinees (Lord, 1980; Wainer, 2000), has therefore been transferred to LLMs: item response theory (IRT) models fitted on the item-level verdicts of previously evaluated systems are used to

pick anchor items (Vivek et al., 2024; Maia Polo et al., 2024), to distill benchmarks (Kipnis et al., 2025), or to administer items adaptively by maximum Fisher information (Hofmann et al., 2025; Li et al., 2026; Zheng et al., 2026; Zhou et al., 2026).

These methods calibrate on hundreds to thousands of systems and then treat the item parameters as known. Many practical settings have only a handful of calibration systems: a new or private benchmark, a freshly generated item pool, a domain where only a few open models can be run, or an agentic benchmark where every evaluation is costly. With four to eight calibrators, item parameters—especially discriminations—are poorly identified, and adaptive selection *capitalizes on calibration error*: it preferentially picks the items whose informativeness is most overestimated (van der Linden and Glas, 2000; Patton et al., 2013). Psychometrics knows remedies: expected response functions that integrate the item-parameter posterior (Mislevy et al., 1994) and fully Bayesian adaptive testing (van der Linden and Ren, 2020; Fink et al., 2025). What is not known is how much they matter in the small-panel regime of LLM evaluation, against which competitors, and with what consequences for the uncertainty that an adaptive test reports.

We study this question with a calibration-marginal adaptive test, EOST-M: items enter the test through their expected response function under an empirical-Bayes item prior, are selected by expected information gain (EIG), and are graded by an independent oracle. We show that this model is exact given the calibrators, that a Gaussian difficulty posterior caps an item's Fisher information whatever the response link, and how entropy, Fisher information and EIG relate under calibration uncertainty (Section 4). Two studies with analysis plans frozen before execution—the second with one target per family cluster and the strongest competitors we could identify—lead to three messages.

1. **Small-panel calibration uncertainty is the real problem.** With eight calibrators, plug-in Fisher-information CAT predicted held-out verdicts worse than EOST-M on every benchmark in both studies (study 2: median 5.9%, Holm-significant on 5 of 8), and its 90% intervals covered the benchmark score only 0.59 of the time.

2. **Accounting for that uncertainty fixes much of it.** Integrating the item posterior (EOST-M) predicted better than constraining the discriminations (a regularized Rasch-type test; median 2.4%, significant on 5 of 8), Bayesian CATs that also integrate the calibrators' abilities were not significantly different from EOST-M (equivalence was not shown), and EOST-M's score error was lower than that of random items under the same model, ATLAS-style CAT, tinyBenchmarks and MINCE on every benchmark, conclusively by the frozen criterion only against MINCE.

3. **Once uncertainty is handled, the information criterion matters much less than first hypothesized.** The information criteria performed almost identically, and EIG beat predictive entropy on held-out prediction by a median of only 1.9%. What remains unsolved is calibrated uncertainty: EOST-M's intervals covered 0.74, and recalibration from the panel raised this to 0.86 without meeting the frozen criterion.

All evidence replays recorded verdicts. Every number in the paper is generated from hashed artifacts by a Docker-only pipeline over pinned public data, and its qualitative statements about results are re-checked automatically against these artifacts.

2 Related work

Calibration error in adaptive testing. Treating estimated item parameters as true inflates measurement precision and biases ability estimates, and adaptive selection amplifies the problem because it seeks the items that look most informative (van der Linden and Glas, 2000; Olea et al., 2012; Patton et al., 2013). Remedies propagate parameter uncertainty into scale scores (Zhang et al., 2011; Yang et al., 2012), replace the item response function by its expectation over the parameter posterior—the expected response function (Mislevy et al., 1994), which we use—or treat all parameters in a fully Bayesian way. van der Linden and Ren (2020) sample the joint posterior of ability and item parameters during the test from stored calibration draws, and Fink et al. (2025) found this approach, with selection by Bayesian maximum information, the most accurate correction in simulated 2PL tests with 100–500 calibration responses per item. We do not claim to be the first to integrate calibration uncertainty into adaptive testing: Lemma 1 shows that our calibration-marginal model is the fully Bayesian model with the calibrators' abilities and the hyperprior held at point estimates, and study 2 compares the two directly.

Item selection. Classical CAT maximizes Fisher information at the current ability estimate (Lord, 1980); Bayesian variants weight it by the ability posterior or use Kullback–Leibler information (Owen, 1975; van der Linden, 1998; Chang and Ying, 1996; Veerkamp and Berger, 1997), and the criteria differ mainly early in a test (Chen et al., 2000). Mutual-information selection appears in classification and multidimensional CAT (Weissman, 2007; Wang and Chang, 2011), as Lindley's information in experimental design (Lindley, 1956; Chaloner and Verdinelli, 1995; Rainforth et al., 2024) and as BALD in active learning (Houlsby et al., 2011), whose motivation—entropy conflates epistemic and aleatoric uncertainty—is our Proposition 2; uncertainty sampling (Lewis and Gale, 1994) and expected error reduction (Roy and McCallum, 2001) are the counterparts of our entropy and score-targeted criteria.

IRT and reduced evaluation for LLMs. IRT has been used to build evaluation scales (Lalor et al., 2016), to weight leaderboard examples (Rodriguez et al., 2021), to select anchors and compressed benchmarks (Vivek et al., 2024; Maia Polo et al., 2024; Kipnis et al., 2025) and to amortize calibration with difficulty predictors (Truong et al., 2025). tinyBenchmarks choose about 100 anchors by clustering item embeddings from hundreds of models and combine them with IRT predictions (gp-IRT); ATLAS (Li et al., 2026) and Fluid Benchmarking (Hofmann et al., 2025) run Fisher-information CAT calibrated on thousands and on about one hundred models; MINCE (Das et al., 2026) uses a few calibration models only to size a random subset. Further work addresses continuous scores, prompt sensitivity, dataset linking, neural and medical item banks and Laplace-approximated neural IRT (Balkır et al., 2026; Jo and Wilson, 2025; Habba et al., 2026; Zhou et al., 2026; Zheng et al., 2026; Mandujano Reyes, 2026), argues for psychometric testing of AI systems (Zhuang et al., 2025), finds item-level IRT inferences unreliable with few calibration models (Jiang et al., 2026)—the regime of this paper—and quantifies error bars for LLM evaluation (Madaan et al., 2024; Miller, 2024).

What is new. The ingredients are classical. We contribute their systematic study in the regime that LLM evaluation faces (panels of 4 to 64 systems, eight real benchmarks from five domains, independent targets), theory for this regime, comparisons under frozen analysis

plans with the competitors that matter there, and an account of uncertainty quantification, recalibration and cost.

3 Calibration-marginal adaptive testing

Items, oracles and verdicts. A test item i is a task with an oracle O_i that maps a system's output to a verdict $y_i \in \{0, 1\}$: exact numeric match, hidden unit tests, or a log-likelihood comparison of answer options. The framework (EOST, for entropy-oracled information-optimal self-testing, optimal in the one-step sense of Section 4) has one structural rule: the system under evaluation may propose items but is never the oracle of its own verdicts, which avoids the circularity of models judging models (Zheng et al., 2023). All items here come from fixed banks; self-generated items were explored only in a historical pilot and are not part of the evidence. Verdicts are deterministic (one recorded evaluation per system and item) or recorded aggregates (a pass rate over samples binarized at one half). Figure 1 summarizes the test.

Measurement model and calibration. Let target s have competence θ_s and respond according to the two-parameter logistic (2PL) model

$$\Pr(y_{si} = 1 \mid \theta_s, a_i, b_i) = \sigma(a_i(\theta_s - b_i)), \qquad a_i > 0. \tag{1}$$

Item parameters $\psi_i = (\log a_i, b_i)$ are calibrated on a *panel* $\mathcal{C}$ of N other systems that excludes the target and every related system (its heuristic family in study 1, its strict cluster in study 2; Section 5). Calibration has four steps shared by every policy: (i) joint maximum a posteriori (MAP) estimation of the calibrators' abilities $\hat{\theta}_{\mathcal{C}}$ and the item parameters, with $\theta_j \sim \mathcal{N}(0, 1)$, $\log a_i \sim \mathcal{N}(0, 1)$, $b_i \sim \mathcal{N}(0, 2^2)$; (ii) an empirical-Bayes hyperprior $\log a_i \sim \mathcal{N}(\mu_a, \sigma_a^2)$, $b_i \sim \mathcal{N}(\mu_b, \sigma_b^2)$ that maximizes the marginal likelihood of all items given $\hat{\theta}_{\mathcal{C}}$ (EM on a 14×65 grid), well determined even for $N = 4$ because it pools hundreds of items; (iii) plug-in parameters $(\hat{a}_i, \hat{b}_i)$, the conditional MAP under that hyperprior; (iv) the exact grid posterior $\pi_i(\psi) \propto p(\psi) \prod_{j \in \mathcal{C}} p(y_{ji} \mid \hat{\theta}_j, \psi)$ on 26×129 points of $(\log a, b) \in [-2.5, 2.5] \times [-8, 8]$. The **plug-in** model uses $P_i(\theta) = \sigma(\hat{a}_i(\theta - \hat{b}_i))$, as in classical CAT and recent LLM work (Hofmann et al., 2025; Li et al., 2026); the **marginal** model uses the expected response function (ERF)

$$\bar{P}_i(\theta) = \int \sigma(a(\theta - b))\, \pi_i(d\psi) \tag{2}$$

(Mislevy et al., 1994), which is the exact response probability given $\hat{\theta}_{\mathcal{C}}$ and the hyperprior (Lemma 1). The two models share $\hat{\theta}_{\mathcal{C}}$ and the hyperprior and differ only in plugging in a mode versus integrating the item posterior.

Item selection. The target's ability posterior π_t is kept on a 201-point grid over $[-5, 5]$ with prior $\mathcal{N}(0, 1)$. At step t an item is chosen by Random (uniform choice), Entropy (largest predictive entropy), MFI (largest Fisher information at the posterior mean), PWFI (posterior-weighted Fisher information; van der Linden, 1998), EIG (largest expected information gain $I(\theta; y_i \mid \mathcal{D}_t)$) or EVR (largest expected reduction of the posterior variance of the unobserved bank total, Proposition 7), each under the plug-in (`:pi`) or the marginal (`:mg`) model (PWFI under the marginal model only). The proposed method, **EOST-M**, is EIG:mg; the state of practice is MFI:pi; the model-free baseline Naive samples

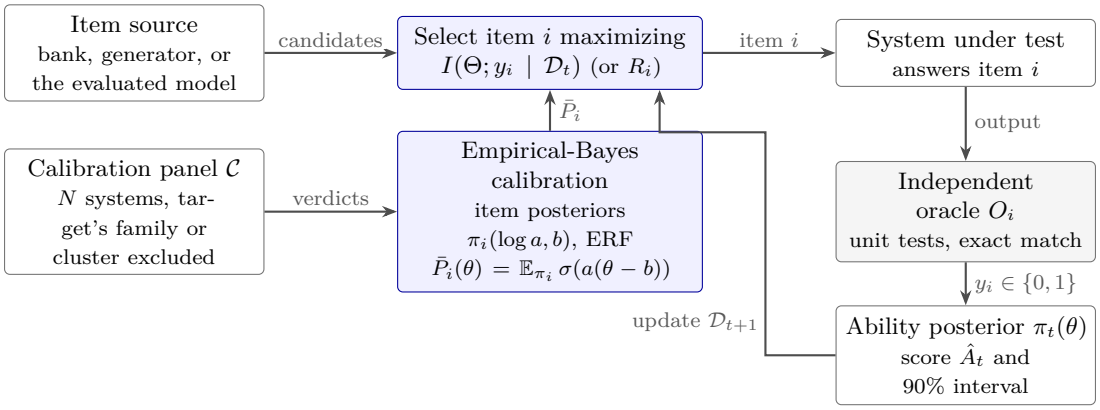


Figure 1: EOST-M: calibration-marginal adaptive testing with an independent oracle.

items uniformly and reports the sample mean with a Wilson interval and finite-population correction. Entropy, EIG and PWFI are matrix–vector products with precomputed item-by-grid matrices, EVR uses the closed form of Proposition 7 on the grid, and ties are broken at random.

Score, intervals and recalibration. The full-bank accuracy $A = \frac{1}{I}\sum_i y_{si}$ is estimated by $\hat{A}_t = \frac{1}{I}\big(\sum_{\text{asked}} y_{si} + \sum_{\text{not asked}} \mathbb{E}_{\pi_t} P_i(\theta)\big)$. Its 90% interval uses the exact mean and variance of the predictive distribution of the unobserved total—a mixture over the ability grid of Poisson-binomial sums—with a normal approximation. Because a model-based interval ignores misspecification and dependence between systems, we also study a leave-one-calibrator-out recalibration ("+R"): for every calibrator c, the item posteriors are recomputed without c's cluster, the same test is replayed on c, whose score is known, and the recalibrated half-width at step t is the 90% quantile of the absolute errors pooled over calibrators and over the steps within a factor 1.5 of t. With $N \le 8$ pseudo-targets this is a cross-validated correction, not a conformal guarantee (Vovk et al., 2005); the rule was chosen among three variants on development banks before study 2.

Endpoints. Competence is a model construct whose scale is fixed only by the prior, so the endpoints are observable. (1) *Held-out predictive log loss*: a holdout of 20% of the bank, drawn once per target and replicate, never administered and shared by all policies and panel sizes, is predicted after each step with $\mathbb{E}_{\pi_t} P_i(\theta)$, and the log loss is averaged over the steps $t \le T$; this strictly proper score rewards both accuracy and calibration. (2) *Score error*: the integrated absolute error (IAE), the average of $|\hat{A}_t - A|$ over $t \le T$. (3) *Coverage* of the 90% interval and the *interval score* $(u - l) + \frac{2}{\alpha}(l - A)_+ + \frac{2}{\alpha}(A - u)_+$ with $\alpha = 0.1$, which penalizes both width and misses (Gneiting and Raftery, 2007). Secondary endpoints include the Brier score, the number of items after which the score error stays below a tolerance, and, on synthetic banks, the error and coverage of the true θ.

4 Theory: what calibration uncertainty does to an adaptive test

Let π_t be the target's ability posterior after t items, with mean μ_t and variance σ_t^2, let $h(p) = -p\log p - (1-p)\log(1-p)$ be the binary entropy, and let P_i be the response function

used for item i (plug-in, or $\bar{P}_i$ of Eq. (2)). All optimality statements are myopic: they concern the next item, not a whole test, for which greedy guarantees require additional structure (Golovin and Krause, 2011).

Lemma 1 (Exactness given the panel's abilities) *Let the calibrators' abilities θ_C and the item hyperparameters η be fixed. Then the item parameters are independent a posteriori given the panel's responses, and for any set S of items answered by a target with ability θ,*

$$p(y_S \mid \theta, \mathcal{D}_{\text{panel}}, \theta_C, \eta) = \prod_{i \in S} \bar{P}_i(\theta)^{y_i} \{1 - \bar{P}_i(\theta)\}^{1-y_i}, \qquad \bar{P}_i(\theta) = \mathbb{E}\big[P_{\psi_i}(\theta) \mid \mathcal{D}_{\text{panel}}, \theta_C, \eta\big]. \tag{3}$$

Integrating θ_C and η as well replaces this product by its expectation over the posterior of (θ_C, η), which no longer factorizes.

Proof Given (θ_C, η), the posterior of the item parameters is proportional to $\prod_i p(\psi_i \mid \eta) \prod_n p(y_{ni} \mid \theta_n, \psi_i)$, a product of single-item factors. The target's responses are conditionally independent given (θ, ψ) and each involves one item, so the integral over ψ factorizes into $\prod_{i \in S} \int p(y_i \mid \theta, \psi_i)\, \pi_i(d\psi_i)$. Averaging over (θ_C, η) afterwards averages products, which factorize only if that posterior is a point mass. ■

EOST-M's ability posterior, predictive distribution and information criteria are therefore exact Bayesian quantities conditional on point estimates of θ_C and η. Fully Bayesian CATs (van der Linden and Ren, 2020; Fink et al., 2025) also integrate (θ_C, η) by Markov chain Monte Carlo (MCMC). When the sampler draws stored item parameters independently for each administered item, as in van der Linden and Ren (2020), the target's posterior uses Eq. (3) with item marginals that also average over (θ_C, η); we call this the *posterior-draw marginal* (PDM) Bayesian CAT, and the version that keeps the product inside the expectation the *joint* fully Bayesian CAT. How much panel-level uncertainty matters is thus an empirical question, answered in Section 7.

Proposition 2 (Entropy–information decomposition) *For every item i, with $\bar{p}_i = \mathbb{E}_{\pi_t} P_i(\theta)$,*

$$U_{\text{EIG}}(i) = I(\theta; y_i \mid \mathcal{D}_t) = \underbrace{h(\bar{p}_i)}_{U_{\text{H}}(i)} - \mathbb{E}_{\pi_t}\big[h(P_i(\theta))\big], \tag{4}$$

so that $0 \le U_{\text{EIG}}(i) \le U_{\text{H}}(i)$, with equality on the right if and only if the item is deterministic given θ, π_t-almost surely.

Proof $I(\theta; y_i \mid \mathcal{D}_t) = H(y_i \mid \mathcal{D}_t) - H(y_i \mid \theta, \mathcal{D}_t)$, where $y_i \mid \mathcal{D}_t$ is Bernoulli$(\bar{p}_i)$ and $y_i \mid \theta$ is Bernoulli$(P_i(\theta))$. Under the marginal model the latter holds with $P_i = \bar{P}_i$, because ψ_i enters only item i's likelihood and its panel posterior does not depend on the target's other responses (Lemma 1). ■

Predictive entropy thus over-values items whose outcome is uncertain even for a known competence: the subtracted term is that irreducible noise.

Proposition 3 (Local equivalence with Fisher information) *Let P_i be three times continuously differentiable with bounded derivatives near μ_t, $P_i(\mu_t) \in (0,1)$, and let $\sigma_t \to 0$ with $\mathbb{E}_{\pi_t}|\theta - \mu_t|^3 = o(\sigma_t^2)$. Then*

$$U_{\mathrm{EIG}}(i) = \tfrac{1}{2}\,\sigma_t^2\,\mathcal{I}_i(\mu_t) + o(\sigma_t^2), \qquad \mathcal{I}_i(\theta) = \frac{P_i'(\theta)^2}{P_i(\theta)\{1 - P_i(\theta)\}}, \tag{5}$$

while $U_{\mathrm{H}}(i) = h\big(P_i(\mu_t)\big) + O(\sigma_t^2)$.

Proof With $\delta = \theta - \mu_t$ and derivatives at μ_t, $P_i(\theta) = P + P'\delta + \frac{1}{2}P''\delta^2 + O(|\delta|^3)$, so $\bar{p}_i = P + \frac{1}{2}P''\sigma_t^2 + o(\sigma_t^2)$. Since h is smooth on $(0,1)$, $h(\bar{p}_i) = h(P) + \frac{1}{2}h'(P)P''\sigma_t^2 + o(\sigma_t^2)$ and $\mathbb{E}\,h(P_i(\theta)) = h(P) + \frac{1}{2}h'(P)P''\sigma_t^2 + \frac{1}{2}h''(P)P'^2\sigma_t^2 + o(\sigma_t^2)$; the region $|\delta| > \varepsilon$ contributes $O(\mathbb{E}|\delta|^3/\varepsilon^3) = o(\sigma_t^2)$. Subtracting and using $h''(p) = -1/\{p(1-p)\}$ gives the claim. ■

Corollary 4 (Entropy is blind to discrimination) *Under the 2PL, the information of item i is $\mathcal{I}_i(\mu) = a_i^2 P_i(\mu)\{1 - P_i(\mu)\}$. Two items with the same $P_i(\mu_t)$ are tied by predictive entropy, while their information ratio is $(a_1/a_2)^2$; under a common discrimination, entropy, maximum Fisher information and EIG induce the same ranking to leading order.*

Proof $P_i' = a_i P_i(1 - P_i)$ gives $\mathcal{I}_i$. With a common a, both $\mathcal{I}_i$ and $h(P_i)$ are strictly decreasing functions of $|P_i - \frac{1}{2}|$. ■

Under a common discrimination, entropy sampling is therefore optimal by accident. Proposition 3 also implies that EIG, posterior-weighted and maximum Fisher information agree to first order, so their differences are early-test phenomena (Chen et al., 2000). The decisive difference comes from calibration uncertainty.

Proposition 5 (Information cap under calibration uncertainty) *Let G be a continuous distribution function (logistic for the 2PL, Φ for the normal ogive), let $P_\psi(\theta) = G\big(a(\theta - b)\big)$, and let the difficulty given the panel be $b \mid a \sim \mathcal{N}(\bar{b}, s^2)$. Then for every θ and every a, the Fisher information of a response under the expected response function satisfies*

$$\bar{\mathcal{I}}(\theta) = \frac{\bar{P}'(\theta)^2}{\bar{P}(\theta)\{1 - \bar{P}(\theta)\}} \le \frac{2}{\pi s^2}, \tag{6}$$

and the bound is attained as $a \to \infty$. For the normal ogive, $\bar{P}(\theta) = \Phi\big(a^{\mathrm{eff}}(\theta - \bar{b})\big)$ with $a^{\mathrm{eff}} = a/\sqrt{1 + a^2 s^2} < 1/s$.

Proof Let $L \sim G$ be independent of b. Since $G(a(\theta - b)) = \Pr(L/a + b \le \theta \mid b)$, a response can be written $y = \mathbf{1}\{L/a + b \le \theta\}$. The law of L does not depend on θ, so the augmented observation (y, L) carries the information $\mathbb{E}_L[\mathcal{I}_{y|L}(\theta)]$. Given $L = \ell$, y is Bernoulli with mean $\Phi(u)$, $u = (\theta - \ell/a - \bar{b})/s$, whose information $s^{-2}\phi(u)^2/\{\Phi(u)(1 - \Phi(u))\}$ is at most $2/(\pi s^2)$, with equality at $u = 0$. Because y is a statistic of (y, L), its information cannot exceed that of (y, L) (Zamir, 1998). As $a \to \infty$, $\bar{P}(\theta) \to \Phi((\theta - \bar{b})/s)$, whose information at $\bar{b}$ is $2/(\pi s^2)$; for the normal ogive, $L/a + b \sim \mathcal{N}(\bar{b}, a^{-2} + s^2)$. ■

The bound is exact for the logistic link (a unit test also checks it numerically), and the Gaussian posterior enters only through the conditional item given L: any difficulty posterior with distribution function F and density f gives $\bar{\mathcal{I}} \le \sup_u f(u)^2/\{F(u)(1-F(u))\}$, for instance $\pi^2/(12s^2)$ for a logistic posterior; for our discrete grid posteriors the cap indicates an order of magnitude. Under the plug-in model an item carries $\hat{a}_i^2/4$ at its difficulty, unbounded in $\hat{a}_i$, and the items that a small panel separates perfectly have both an inflated $\hat{a}_i$ and a large s_i: plug-in selection concentrates on them—the capitalization on calibration error (van der Linden and Glas, 2000; Patton et al., 2013)—whereas the expected response function caps their information whatever $\hat{a}_i$ is. A shrinkage prior on a_i limits the inflation of $\hat{a}_i$, but not the uncertainty s_i about where the item sits.

Corollary 6 (Averaging information is not integrating uncertainty) *For any item-parameter posterior, $\bar{\mathcal{I}}_i(\theta) \le \mathbb{E}_{\pi_i}[\mathcal{I}_{i,\psi}(\theta)]$.*

Proof $\mathcal{I} = P'^2/P + P'^2/(1-P)$ is jointly convex in (P, P'), and $(\bar{P}_i, \bar{P}'_i)$ is the posterior average of $(P_{i,\psi}, P'_{i,\psi})$; Jensen's inequality concludes. ■

Criteria that average Fisher information over parameter uncertainty—the Bayesian maximum-information criterion of fully Bayesian CAT (van der Linden and Ren, 2020; Fink et al., 2025) and Laplace-approximated variants (Mandujano Reyes, 2026)—can therefore be optimistic about poorly calibrated items and do not inherit the cap of Proposition 5.

Myopic score-targeted design. Practitioners often care about the full-bank accuracy rather than about θ. Let U be the set of unobserved items and $S_U = \sum_{k \in U} y_k$; the score-targeted criterion (EVR) asks the item that minimizes the expected posterior variance of S_U after one more response.

Proposition 7 (One-step score-targeted design) *For an unobserved item $i \in U$, let $v_i = \bar{p}_i(1-\bar{p}_i)$, let $M_{-i}(\theta) = \sum_{k \in U \setminus \{i\}} P_k(\theta)$, and let $c_i = \mathrm{Cov}_{\pi_t}(P_i(\theta), M_{-i}(\theta))$. The expected one-step variance reduction obtained by asking i is exactly*

$$R_i = \mathrm{Var}(S_U \mid \mathcal{D}_t) - \mathbb{E}_{y_i}[\mathrm{Var}(S_{U\setminus\{i\}} \mid \mathcal{D}_t, y_i)] = \frac{(v_i + c_i)^2}{v_i}. \tag{7}$$

Under the conditions of Proposition 3, and if $P'_i(\mu_t)M'_{-i}(\mu_t) \neq 0$, $c_i = \sigma_t^2 P'_i(\mu_t)M'_{-i}(\mu_t)\{1 + o(1)\}$ and $R_i = v_i + 2c_i + \sigma_t^4 M'_{-i}(\mu_t)^2\,\mathcal{I}_i(\mu_t)\{1 + o(1)\}$.

Proof Write $S_U = y_i + S_{-i}$. By the law of total variance, $R_i = \mathrm{Var}(y_i) + 2\,\mathrm{Cov}(y_i, S_{-i}) + \mathrm{Var}(\mathbb{E}[S_{-i} \mid y_i])$, with all moments conditional on $\mathcal{D}_t$. Here $\mathrm{Var}(y_i) = v_i$; conditional independence given θ gives $\mathrm{Cov}(y_i, S_{-i}) = c_i$; and since y_i is binary, $\mathbb{E}[S_{-i} \mid y_i]$ is affine in y_i with slope c_i/v_i, so the last term is c_i^2/v_i. First-order expansions of P_i and M_{-i} around μ_t give c_i, the remainders contributing $O(\mathbb{E}|\delta|^3 + \mathbb{E}\delta^4) = o(\sigma_t^2)$ on the bounded ability grid, and c_i^2/v_i then gives the last term. ■

The first term of R_i is a monotone transform of predictive entropy; the last is Fisher information weighted by the squared sensitivity of the remaining bank to θ. Score-targeted

selection therefore behaves like information sampling while competence is uncertain and like uncertainty sampling once it is pinned down, and because M'_{-i} grows with the number of unobserved items, the information regime lasts longer on larger banks. "Information-optimal" is thus relative to an estimand and to a one-step horizon.

5 Experimental design

Data. Table 1 describes the eight confirmatory banks of both studies and the three development banks used only to design the protocols. Open LLM Leaderboard (Beeching et al., 2023) verdicts for 395 models on GSM8K (Cobbe et al., 2021), ARC-Challenge (Clark et al., 2018), HellaSwag (Zellers et al., 2019), WinoGrande (Sakaguchi et al., 2020), MMLU (Hendrycks et al., 2021) and TruthfulQA (Lin et al., 2022) come from the release of Maia Polo et al. (2024) (HellaSwag and MMLU subsampled to 2,000 items, TruthfulQA's MC2 score binarized at 0.5); SWE-bench Verified and Lite (Jimenez et al., 2024) verdicts (instance resolved by the official harness) come from the public submissions (175 and 84 systems); LiveCodeBench (Jain et al., 2025) pass rates of 28 models and CRUXEval (Gu et al., 2024) pass rates of 31 models are binarized at one half. In the table, J, F and I count systems, heuristic families and items, a "single" verdict is one deterministic evaluation per system and item, an "aggregate" verdict a binarized pass rate, and the last column counts the items that no system solves. Every source is pinned to a git commit and every processed matrix is hashed. Three synthetic banks with known abilities (300 systems and 600 items each: a well-specified 2PL, 60% of items with a guessing floor of 0.25, and half of the items depending on a second ability dimension correlated 0.6 with the first) are always fitted with the unidimensional 2PL and never enter a confirmatory decision.

Table 1: Response matrices of the eight confirmatory and three development banks.

Bank	Role	Domain	Oracle	Verdict	J	F	I	Accuracy	Unsolved
GSM8K	1, 2	math	numeric match	single	395	227	1319	0.00–0.77	13
ARC-C	1, 2	multiple choice	log-lik. choice	single	395	227	1172	0.26–0.73	51
HellaSwag	1, 2	multiple choice	log-lik. choice	single	395	227	2000	0.31–0.90	43
WinoGrande	1, 2	multiple choice	log-lik. choice	single	395	227	1267	0.59–0.87	1
MMLU	1, 2	multiple choice	log-lik. choice	single	395	227	2000	0.26–0.80	0
SWE-bench V.	1, 2	agentic SE	repository tests	single	175	123	500	0.00–0.79	26
LiveCodeBench	1, 2	code generation	hidden tests	aggregate	28	15	1055	0.23–0.87	46
CRUXEval-O	1, 2	code reasoning	execution	aggregate	31	12	800	0.22–0.78	98
SWE-bench Lite	dev.	agentic SE	repository tests	single	84	49	300	0.00–0.60	35
CRUXEval-I	dev.	code reasoning	execution	aggregate	31	12	800	0.14–0.78	59
TruthfulQA	dev.	multiple choice	log-lik. choice	aggregate	395	227	817	0.30–0.73	47

Families and strict clusters. Systems that share an underlying model are not independent. *Heuristic families* group systems by organization namespace (Open LLM Leaderboard), by model line (LiveCodeBench, CRUXEval) or by shared underlying LLM (SWE-bench). Because they miss renamed copies, merges and cross-organization fine-tunes, study 2 uses *strict clusters*: connected components of the graph that joins two systems if they share a family or if their standardized 2PL residuals correlate at least 0.5, pooled over the banks that share systems. There are 176 strict clusters on the Open LLM Leaderboard (the largest with 39 systems), 92 on SWE-bench Verified, 15 on LiveCodeBench and 12 on CRUXEval.

Study 1. For each bank, up to 60 targets were drawn by accuracy decile (all systems for LiveCodeBench and CRUXEval). For each target and replicate $r = 1, \ldots, 5$ a holdout was drawn, and for each panel size $N \in \{4, 8, 16, 32, 64\}$ a panel outside the target's heuristic family. The twelve policies of Section 3 (five criteria under both models, PWFI under the marginal model, and Naive) were replayed on the target's recorded verdicts with a budget of $T = 100$ items. At the primary panel size $N = 8$, **H1** states that EIG:mg has a lower held-out log loss than Entropy:mg, **H2** the same against MFI:pi, **H3** that EIG:mg has a lower score error than Naive, and **H4** that its 90% intervals are better calibrated than MFI:pi's ($|\text{coverage} - 0.9|$). Each contrast is a two-sided Wilcoxon signed-rank test (Wilcoxon, 1945) on per-target values, with Holm correction (Holm, 1979) over the whole family; a hypothesis is *supported* if the adjusted test favours EOST-M on at least five of the eight banks and the baseline on none. The protocol was designed on the development banks, where fixed weakly-informative priors shrank unsolved items towards an arbitrary difficulty, so empirical-Bayes hyperpriors were adopted for every policy. Plan, code and data manifests were then hashed and committed before any confirmatory run; as no external registry was used, we speak of a *frozen analysis plan* rather than a preregistration (Nosek et al., 2018).

Study 2. Designed after an external review of study 1 and frozen in the same way, study 2 uses one target per strict cluster (at most 40 per bank, stratified by accuracy) and panels ($N \in \{4, 8, 16\}$, three replicates) that exclude the target's whole cluster. It adds the strongest competitors we could identify: a regularized Rasch-type plug-in CAT (MFI:1PR: one common discrimination with $\log a \sim \mathcal{N}(0, 0.05^2)$) and its marginal version (EIG:1PRMG); the PDM Bayesian CAT, a Gibbs sampler for the panel's hierarchical 2PL with a translation move for the overall location and eight retained draws, selecting by Bayesian maximum information as recommended by Fink et al. (2025) or by EIG; RANDOM:MG, random items scored by EOST-M's model, which isolates the value of adaptivity; an ATLAS-style 3PL CAT (Li et al., 2026) with the lower asymptote at the chance rate and a random choice among the five most informative items (Kingsbury and Zara, 1989); tinyBenchmarks (K-Means anchors on the panel's item embeddings and the gp-IRT estimator; Maia Polo et al., 2024) and MINCE (Monte Carlo sizing and the mean of a frozen random subset; Das et al., 2026) at budgets of 25, 50 and 100 items; MFI:pi and Naive; and two empirical-Bayes variants of EOST-M (a cross-fitted hyperprior and a Student-t family). Eleven superiority hypotheses at $N = 8$, with Holm correction over 11×8 tests and the same decision rule, replicate H2–H4 (R2–R4) and compare EOST-M with the Rasch-type plug-in (C1) and marginal (C2) tests on log loss, with Random:mg (C4) and the ATLAS-style CAT (C7) on score error, with tinyBenchmarks (C8) and MINCE (C9) on the score error after 50 items, and the recalibrated interval with the raw one (C5b) and with Naive's Wilson interval (C6) on the interval score. Two further decisions test the equivalence of the PDM Bayesian CAT and EOST-M within ±1% (C3: 90% bootstrap interval inside the margin on at least six banks) and the calibration of the recalibrated interval (C5a: mean coverage in $[0.85, 0.95]$ on at least six banks). A temporal component on SWE-bench Verified uses the latest quarter of submissions as targets and only earlier submissions as calibrators.

Deviations and reproducibility. No deviation affected a confirmatory computation. While study 2 was running, before any result was inspected, an independent review of its code found that tinyBenchmarks' K-Means can return fewer anchors than its budget and

that the sampler's frozen burn-in (10 sweeps) was too short on some banks; the frozen run was completed unchanged, every frozen decision is reported as computed, and the post hoc corrections, specified before the results were seen, are reported alongside. Analyses outside the frozen plans are labelled exploratory or post hoc, with decision rules recorded before running. Everything runs in a pinned CPU-only Docker image without network access (study 1: 172,620 adaptive tests, about 3.8 h on two cores; study 2: about 3 h).

6 Study 1: calibration-marginal versus plug-in adaptive testing

Frozen decisions. Table 2 reports the four contrasts at $N = 8$ as the relative improvement of EOST-M over the baseline of each hypothesis (log loss for H1 and H2, score error for H3, $|\text{coverage} - 0.9|$ for H4); an asterisk marks a Holm-significant difference, and none favoured a baseline. **H2 is supported**: EOST-M lowered the held-out log loss on 8 of 8 benchmarks (median 5.2%), significantly on 7; the median gain was 7.1% at $N = 4$ and only 1.1% at $N = 64$. **H3 is supported**: the score error fell by a median of 54.4% relative to naive subsampling (7 benchmarks significant). **H4 is supported**: EOST-M reduced the miscalibration of plug-in Fisher CAT by a median of 44.2% (7 significant). **H1 is not supported**: EIG beat entropy on 8 of 8 benchmarks, but by a median of only 1.9% and significantly on 4, short of the threshold; the sign predicted by the EOST thesis holds at every panel size, but its size on held-out prediction is small once calibration uncertainty is integrated (Corollary 4). On GSM8K, H2–H4 point the same way without significance; its leaderboard spans the whole accuracy range, so an eight-system panel leaves the fewest items unlocated (21% against a median of 39% elsewhere).

Table 2: Study 1: relative improvement (%) of EOST-M for H1–H4 at $N = 8$.

Benchmark	targets	H1 vs Entropy:mg	H2 vs MFI:pi	H3 vs Naive	H4 vs MFI:pi
GSM8K	60	+2.7*	+2.4	+16.8	+11.6
ARC-C	60	+0.4	+4.7*	+61.0*	+41.6*
HellaSwag	60	+5.5*	+9.6*	+68.4*	+46.8*
WinoGrande	60	+3.1*	+4.9*	+53.4*	+57.3*
MMLU	60	+2.9*	+5.0*	+55.3*	+59.0*
SWE-bench V.	60	+0.7	+6.1*	+46.9*	+31.7*
LiveCodeBench	28	+1.0	+6.4*	+51.3*	+48.1*
CRUXEval-O	31	+0.5	+5.4*	+57.7*	+33.4*
decision		4/0 not supported	7/0 supported	7/0 supported	7/0 supported

Dependent targets. Many Open LLM Leaderboard systems are fine-tunes or merges of a few base models. Re-analysed post hoc with heuristic families as units, the frozen decisions are unchanged. With strict clusters (12 to 50 units per benchmark), H3 and H4 remain supported, but H2 is significant on only 4 benchmarks, below the threshold, although its median effect is unchanged (5.4%, 95% cluster-bootstrap interval 4.3–6.4%). The predictive gain is therefore not in doubt, but its significance with targets as units overstates the evidence; study 2 was designed with one target per cluster to settle this.

Where the gain comes from. EOST-M differs from plug-in Fisher CAT in its criterion and in its measurement model, and Table 3 crosses the two. At $N = 8$, replacing plug-in parameters by expected response functions improves held-out log loss by a median of +4.4% even with entropy selection, whereas changing the criterion under plug-in parameters changes little (−1.8% for entropy, +0.0% for EIG); combining both gives +5.2%. For the score the criterion matters more: entropy is the worst adaptive rule under both models (−9.2% under the marginal model), whereas EIG:mg improves the score IAE by +7.5%. Under the marginal model the four information criteria (MFI, PWFI, EIG, EVR) are practically indistinguishable—their mean log losses differ by at most 0.1% and their score errors by at most 3%—as the first-order equivalence of Proposition 3 predicts. Most of the gain thus comes from integrating calibration uncertainty, and EIG is the principled member of an equivalence class rather than a uniquely better rule.

Table 3: Study 1: gain over MFI:pi by criterion and measurement model (median, %).

		holdout log loss				score IAE			
N	B	Ent:pi	EIG:pi	Ent:mg	EIG:mg	Ent:pi	EIG:pi	Ent:mg	EIG:mg
4	8	−0.8	−0.1	+5.0	+7.1	−12.3	−0.8	−15.2	+11.7
8	8	−1.8	0.0	+4.4	+5.2	−17.5	+0.7	−9.2	+7.5
16	8	−1.4	0.0	+2.0	+3.6	−18.7	−0.7	−10.9	+5.9
32	6	−1.8	0.0	−0.5	+1.9	−23.4	+0.2	−22.8	+7.4
64	6	−2.0	+0.1	−1.3	+1.1	−17.5	+0.6	−16.7	+6.6

The gain over plug-in Fisher CAT decreases monotonically with the panel size on every benchmark (Figure 2, panel a, where an asterisk marks a Holm-adjusted $p < 0.05$), as Proposition 5 predicts, since the uncertainty of item difficulties shrinks as calibrators are added; panel b gives the score error and panel c the contrast with entropy (H1). Calibration diagnostics show the mechanism. With four calibrators 56% of the items were answered identically by every calibrator (15% with 64), so their difficulty is located only by the prior. Under the empirical-Bayes prior, plug-in discriminations above 4 are practically absent with small panels (0% of items at $N = 4$, 10% at $N = 64$), while the expected response function shrinks the median effective discrimination from 1.99 to 0.98 at $N = 4$, as Proposition 5 prescribes. The overconfidence of plug-in testing at small N therefore comes mainly from treating poorly located difficulties as known, and across benchmarks and panel sizes the log-loss gain of EOST-M grows with the fraction of unlocated items (Spearman $\rho = 0.85$, exploratory).

Learning curves and efficiency. Every model-based policy starts below the naive estimator, because the calibrated model predicts the bank before any item is asked (Figure 3). Plug-in policies reduce the error fastest in the first items (at $t = 5$ a plug-in policy is ahead of EOST-M on 7 of 8 benchmarks) but then level off: between $t = 25$ and $t = 100$ the error of plug-in Fisher CAT falls by a median of 27%, that of EOST-M by 41%, and after 100 items EOST-M is ahead on 8 of 8 benchmarks—the signature of a bias floor inherited from the plug-in parameters. At $N = 8$, EOST-M matched the score error of 100 uniformly sampled items after a median of 22.5 items (between 8 and 70 across benchmarks), and after 8 with the largest panels. Adaptivity itself contributes: random items scored by the

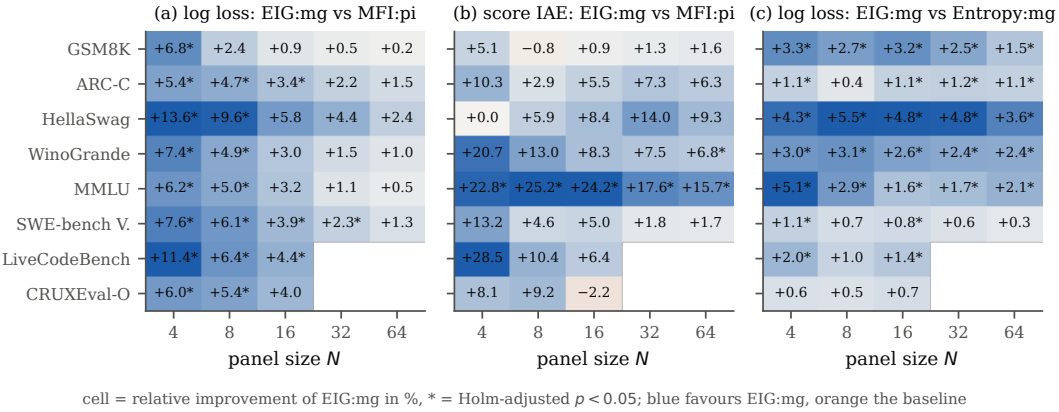


Figure 2: Study 1: relative improvement (%) of EOST-M by benchmark and panel size.

same marginal model had a mean score IAE of 0.0432 against 0.0309 for EOST-M, whose error was significantly lower on 7 of 8 benchmarks. Late in the test, entropy selection under the marginal model catches up, and the crossover comes later on larger banks (Spearman $\rho = 0.73$ over 36 benchmark–panel combinations, exploratory), the regime change described by Proposition 7; the score-targeted criterion moved in the predicted direction, but only part of the way. Short adaptive tests also rank systems well: after 25 items, Kendall's τ with the true accuracy had a median of 0.77 for EOST-M, 0.74 for plug-in Fisher CAT and 0.54 for random items (exploratory).

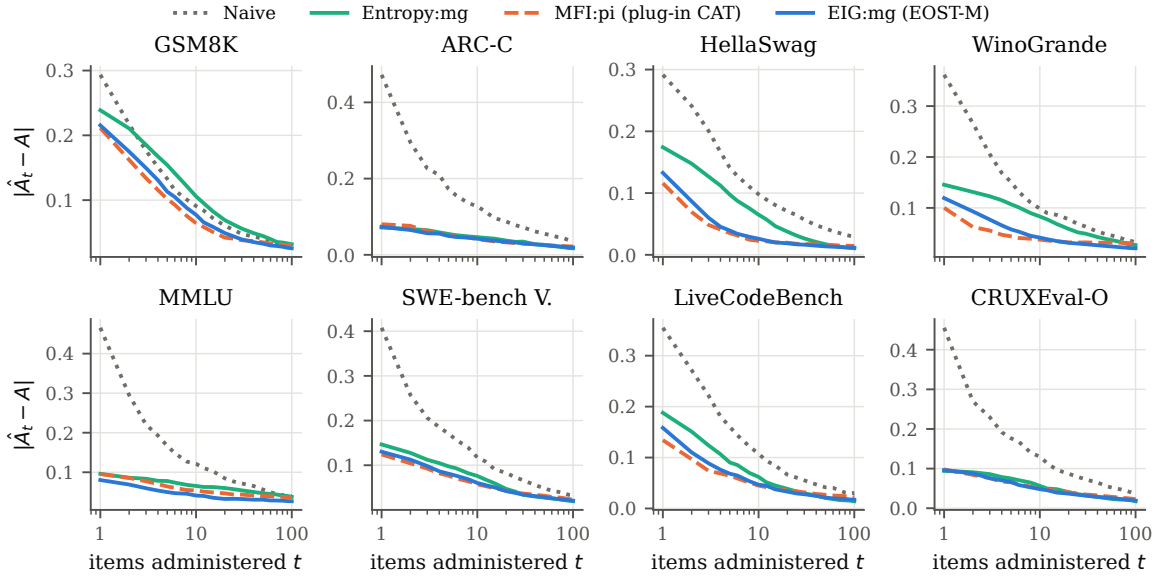


Figure 3: Study 1: absolute score error as items are administered ($N = 8$).

Uncertainty. Plug-in posteriors are overconfident (Figure 4a): at $N = 8$ the 90% interval of plug-in Fisher CAT covered the score 0.60 of the time, EOST-M's 0.75 and Naive's

model-free Wilson interval 0.90. Marginalization removes much of the overconfidence, but no model-based adaptive policy reaches nominal coverage on real data. On the synthetic banks (Figure 4b), EOST-M's coverage at $N = 8$ is 0.89 when the model is well specified, 0.88 with guessing and 0.81 with a second ability dimension, and as the panel grows coverage rises towards the nominal level on synthetic data but falls on real data: more calibrators sharpen the item posteriors, which helps when the model is adequate and exposes its misfit when it is not. With known abilities, EOST-M also had the smallest ability error at $N = 8$ on all three synthetic banks (0.244, 0.391 and 0.469, against 0.251, 0.415 and 0.478 for plug-in Fisher CAT).

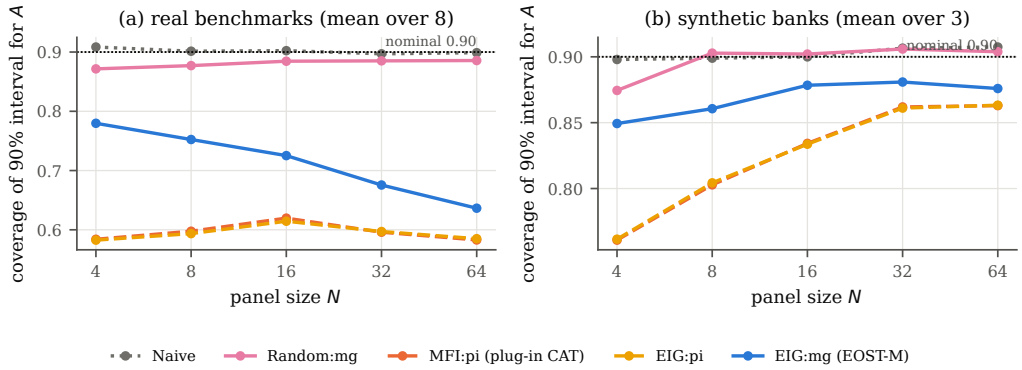


Figure 4: Study 1: coverage of the 90% score interval as a function of the panel size.

Sensitivity and post hoc probes. Removing near-duplicate calibrators leaves the H2 gain essentially unchanged (+6.1% against +4.9% on the same benchmarks), but the empirical-Bayes hyperprior is essential: with a fixed weakly-informative prior the H2 gain falls to +2.8% (main analysis: +5.2%), because for an item that the panel cannot locate the expected response function is essentially the prior predictive. The one-parameter variant of the frozen plan is degenerate—its likelihood is invariant to a joint rescaling, and its mode inflates the common discrimination (median 1.5×10^4)—and is not interpreted. A post hoc variant with $\log a \sim \mathcal{N}(0, 0.05^2)$ confirms Corollary 4: with one common discrimination the EIG–entropy difference is +0.4% under the marginal model and −0.4% under plug-in parameters, against +1.9% and +1.6% with item-specific discriminations.

Three post hoc probes, with decision rules recorded before running, then tested a competing explanation of H2. (i) Posterior-mean plug-in parameters are no remedy (log loss 15.5% worse than EOST-M), but a regularized Rasch-type plug-in test is a strong baseline: EOST-M's log loss was lower on all 8 benchmarks, yet only by a median of 2.1% (significant on 4), and the Rasch-type intervals were better calibrated (coverage 0.81 against 0.75); averaging Fisher information over the item posterior selected items as well as EIG (median difference 0.0%), so the optimism of Corollary 6 did not matter. (ii) Shrinking EOST-M's discriminations towards a common value did not combine the two remedies: no setting met the criterion fixed in advance (coverage between 0.74 and 0.76). (iii) The Rasch-type intervals were wider on all 8 benchmarks without more accurate estimates, and across all cells coverage was almost a monotone function of the ratio of interval width to score error (Spearman $\rho = 0.98$), a ratio below that of a calibrated Gaussian interval in every cell. The

constrained model is better calibrated because it is less confident, and since random items under the same model cover 0.88 of the time, the shortfall arises when adaptive selection meets a misspecified model. Study 2 tests the Rasch-type competitor under a frozen plan.

7 Study 2: independent targets and strong competitors

Study 2 evaluated 267 independent targets (12 to 40 per benchmark). Table 4 lists every frozen decision with the number of benchmarks on which a Holm-significant difference favours EOST-M or the competitor, the number on which EOST-M is better, and the median relative improvement (for C3 the relative difference of the PDM Bayesian CAT, for C5a the range of coverage; C5a, C5b and C6 concern the recalibrated interval, marked by an asterisk). Of the 13 decisions, 6 were supported. Apart from C5b, every superiority contrast favoured EOST-M on at least seven of the eight benchmarks and no competitor was significantly better on any benchmark; four of the five superiority hypotheses that were not supported (C2, C4, C7, C8) missed the threshold by one benchmark. CRUXEval-O, with only 12 independent clusters, reached significance in no contrast.

Table 4: Study 2: frozen hypotheses at $N = 8$ and their decisions.

	EOST-M versus	Endpoint	Holm sig.	positive	median	Supported
R2	plug-in 2PL (MFI:pi)	log loss	5/0	8/8	+5.9	yes
R3	Naive	score IAE	6/0	8/8	+53.9	yes
R4	plug-in 2PL (MFI:pi)	$\lvert\text{cov} - 0.9\rvert$	5/0	8/8	+41.6	yes
C1	Rasch-type plug-in	log loss	5/0	7/8	+2.4	yes
C2	Rasch-type marginal	log loss	4/0	8/8	+1.8	no
C4	Random:mg	score IAE	4/0	8/8	+21.1	no
C5b	raw EOST-M interval*	interval score	0/2	4/8	+1.0	no
C6	Naive (Wilson)*	interval score	6/0	7/8	+35.0	yes
C7	ATLAS-style 3PL CAT	score IAE	4/0	8/8	+26.0	no
C8	tinyBenchmarks	error @50	4/0	8/8	+29.6	no
C9	MINCE	error @50	5/0	8/8	+44.8	yes
C3	PDM Bayesian CAT (equivalence)	log loss	4/8 within ±1%		+0.32	no
C5a	(coverage in $[0.85, 0.95]$)*	coverage	5/8 banks inside		0.81–0.91	no

Replication with independent targets. The study-1 results held: EOST-M's held-out log loss was lower than plug-in Fisher CAT's on every benchmark (R2, median 5.9%, 5 significant), more so with four calibrators (8.3%) than with sixteen (4.6%), and by the same amount for deterministic (6.1%) and aggregate verdicts (5.9%); R3 and R4 were also supported, and plug-in Fisher CAT's intervals covered the score only 0.59 of the time, against 0.74 for EOST-M.

Integration, constraint and Bayesian CATs. EOST-M predicted better than the Rasch-type plug-in test (C1, median 2.4%), more so with sixteen calibrators (4.0%) than with four (2.9%), where a single discrimination becomes a visible misspecification; the integrated one-parameter model sat in between (C2, 1.8%). The two tests had similar score errors (median 1.6%, no benchmark significant), and the Rasch-type intervals were again better calibrated before recalibration (0.81). The PDM Bayesian CAT differed from EOST-M by

a median of 0.32% in log loss and was within ±1% on only 4 benchmarks, so equivalence (C3) was not supported. Its frozen burn-in proved too short on HellaSwag, where the mean calibrator ability drifted by a median of 0.51 during the retained sweeps; with either burn-in (10 or, post hoc, 60 sweeps) it was not significantly different from EOST-M on any benchmark, in log loss or score error (corrected log loss higher by a median of 0.39% with Bayesian maximum information and 0.50% with EIG). The exploratory joint fully Bayesian CAT, which keeps the uncertainty about the location of the calibration shared across items, was the closest to EOST-M with EIG selection (median 0.05%). At $N = 4$ the corrected PDM Bayesian CAT was never significantly better than EOST-M (within ±1% on 4 and 2 benchmarks with the two criteria). With eight calibrators, integrating their abilities and the hyperprior on top of the item parameters thus made no significant difference, but equivalence within ±1% was not shown.

Adaptivity and benchmark reduction. All methods used the same eight calibrators; ATLAS and tinyBenchmarks were designed for hundreds to thousands, so these comparisons concern the small-panel regime, not the methods in their native settings. Under the same calibrated model, adaptive selection reduced the score error of random items by a median of 21.1% (C4, 4 benchmarks significant), more with larger panels (17.2% at $N = 4$, 29.3% at $N = 16$). Table 5 gives the score errors at 25, 50 and 100 items, averaged over targets and benchmarks; tinyBenchmarks and MINCE choose their items without adapting to the target. After 50 items EOST-M's error was 29.6% below tinyBenchmarks' gp-IRT (C8) and 44.8% below MINCE's subset mean (C9, supported), and over the test 26.0% below the ATLAS-style CAT's (C7), whose plug-in 3PL parameters and randomesque selection made it the least accurate model-based policy (coverage 0.48). With eight calibrators, tinyBenchmarks' K-Means cannot always find as many anchors as its budget, because its item embeddings take only as many values as there are distinct response patterns; an audit decided before the results were seen showed that restricting C8 to full-budget cells, or comparing with EOST-M after the number of items tinyBenchmarks actually used, again gave 4 significant benchmarks. MINCE's own stopping rule asked for a median of 158 items and then had a score error of 2.50 points, more than EOST-M after 100 items (2.13).

Table 5: Study 2: absolute score error (points) after 25, 50 and 100 items at $N = 8$.

Method	25 items	50 items	100 items
EOST-M (EIG:mg)	3.39	2.81	2.13
ATLAS-style 3PL CAT	4.46	4.20	3.79
Random:mg	4.78	3.34	2.42
tinyBenchmarks gp-IRT	4.96	4.39	4.49
tinyBenchmarks p-IRT	4.69	3.92	3.31
tinyBenchmarks anchors	6.23	5.62	5.37
MINCE subset mean	6.71	4.69	3.19
Naive (Wilson)	7.05	4.88	3.26

Calibrated uncertainty. Model-based intervals remained too narrow: EOST-M covered the score 0.74 of the time, the PDM Bayesian CAT 0.76, the Rasch-type test 0.81 and plug-in Fisher CAT 0.59, whereas Random:mg reached 0.88 and Naive's Wilson interval 0.90. Recalibration raised EOST-M's coverage to 0.86, inside $[0.85, 0.95]$ on 5 benchmarks, one

short of the criterion (C5a), and did not improve the interval score (C5b; significantly worse on HellaSwag and WinoGrande): the wider intervals cost about as much as the misses they removed. The recalibrated intervals had a much better interval score than Naive's Wilson intervals (C6, median 35.0%) because they were much narrower (mean width 0.134 against 0.235). Recalibration brought the Rasch-type test inside the band on all eight benchmarks, but with intervals about twice as wide (0.271). It needs calibrators as pseudo-targets and helped more as the panel grew (coverage 0.82, 0.86 and 0.88 at $N = 4$, 8 and 16), and mixing one uniformly drawn item in four into the adaptive test raised coverage only to 0.77.

Temporal validation and the hyperprior. On the 31 SWE-bench Verified submissions of the latest quarter, calibrated only on earlier submissions, 10 of the 11 frozen contrasts favoured EOST-M at a Holm-adjusted $p < 0.05$ within the component and none favoured a competitor; EOST-M's log loss was 6.7% lower than plug-in Fisher CAT's and 3.1% lower than the Rasch-type test's. Because the empirical-Bayes hyperprior is essential and estimated from the same panel, it was checked post hoc and varied under the frozen plan. Resampling items moves the hyperparameters by at most 0.07 (standard deviation, $N = 8$), whereas resampling calibrators moves them about as much as the variation between panels: the hyperprior reflects which systems are in the panel, not noise in the items. Prior predictive checks reproduce the spread of the items' proportions correct, but the share of items with identical panel verdicts only on average, and real systems agree with each other less, item by item, than a unidimensional 2PL with independent items implies (mean item–rest correlation 0.39 observed, 0.41 simulated): the response model is misspecified in its dependence structure, one reason why model-based intervals are too narrow. A cross-fitted hyperprior, estimated for each item on the other half of the items, left EOST-M's log loss unchanged (median 0.0%), and Student-t hyperpriors made it slightly worse (0.4%): the gain of EOST-M does not come from fitting the prior to the items it is applied to.

8 Robustness of the study-2 conclusions

A second review asked whether the response-derived clusters favour EOST-M, whether the Bayesian CATs were estimated with enough MCMC, and whether the effects generalize across benchmarks. Three follow-ups (RF1–RF3), whose plans and decision rules were written before they were run, answer these questions, together with exploratory extensions (RF2b–RF2d, RF3b) planned in the same way; none changes a frozen decision.

Clusters that never see the target's responses (RF1). The strict clusters use residuals computed from every system's responses, the target's included, and excluding the calibrators that resemble a target could make its panel harder to calibrate. Keeping the frozen targets, holdouts and seeds, we rebuilt the panels from heuristic families alone, from residuals computed on the other benchmarks, and with residual thresholds from 0.4 to 0.7. Under every rule, R2 and C1 kept their frozen median improvements (5.9% and 2.4%), R2 remained positive on all eight benchmarks with 4–6 significant, and C1 positive on seven with 5 significant and none against: the robustness criterion fixed in advance was met.

MCMC quality of the Bayesian CATs (RF2–RF2d). We ran the frozen sampler with four long chains on 48 panels and computed rank-normalized split $\hat{R}$ and bulk effective sample sizes (Vehtari et al., 2021). The overall location converged ($\hat{R} < 1.05$ on 92% of

panels), but σ_b, μ_a and σ_a did not ($\hat{R} < 1.05$ on at most 8%): multiplying all abilities and difficulties by a constant and dividing the discriminations by it leaves the likelihood unchanged, so with eight calibrators only their $\mathcal{N}(0, 1)$ prior fixes the scale of the calibration, and the frozen sampler has no move along this ridge. Under the rule fixed in advance the study-2 statement about the Bayesian CATs was not confirmed: the long chains changed the cell-level log loss by a median of 0.61–1.02% relative to a burn-in of 60, more than the 0.5% allowed, while the long-chain arms differed from EOST-M by a median of −0.45 to 0.20%, with the benchmarks split. Exploratory samplers then added a Metropolis–Hastings move along the scale ridge with the item parameters summed out (RF2b, a partially collapsed Gibbs sampler with a discrete form of parameter expansion; van Dyk and Park, 2008; Liu and Wu, 1999) and sampled the model of Lemma 1 exactly, with collapsed moves for the hyperparameters, the scale and the location (RF2d). RF2d came much closer to convergence (medians of $\hat{R}$ at most 1.058, largest value 1.25 against 3.04 for the frozen sampler) but reached $\hat{R} < 1.05$ on only 42% of panels for its slowest quantity, short of the 90% fixed in advance, so a fully converged comparator was not obtained. Across all 18 combinations of Bayesian arm and sampler, the median difference from EOST-M in held-out log loss stayed within ±0.45%, with the benchmarks split: the comparison did not move as the sampler improved, which makes an artefact of the frozen chain unlikely. The RF2d arms covered the score as often as EOST-M on the same cells (0.76–0.79 against 0.78), and their score error was lower by a median of 5–11% over cells but on only 4–5 of 8 benchmarks, a possible advantage that 48 cells cannot establish.

Effects across benchmarks (RF3, RF3b). The frozen rule requires significance on five of eight benchmarks after a Holm correction over 88 tests, so an effect can favour EOST-M everywhere and still be declared not supported. A secondary random-effects meta-analysis that treats the benchmarks as independent found pooled improvements excluding zero for 10 of the 11 superiority hypotheses, including the 4 near misses (C2, C4, C7, C8). Five benchmarks, however, share the same 395 Open LLM Leaderboard systems. Counting them as one source (RF3b: 4 sources, with a bootstrap over the shared systems), 6 of the 11 intervals excluded zero (R2, R3, R4, C2, C6, C9), and of the near misses only C2 kept an interval above zero; the dependence through shared systems was small (median correlation 0.02), and the loss comes from having four heterogeneous sources. Most effects point the same way across benchmarks; the analysis does not show that they generalize to a new one.

9 What adaptive evaluation costs

Adaptive testing saves items only after a calibration panel has answered the whole bank. Counting one administered item as the unit of cost, evaluating M new systems costs MI with the full bank of I items and $NI + MT$ with a fresh panel of N calibrators and tests of T items, so adaptive testing pays off once $M > NI/(I - T)$. Table 6 gives, for $N = 8$ and $T = 100$, this break-even point (between 8.4 and 10 new systems), the number of items with which EOST-M matches the score error of 100 items sampled uniformly and scored by the sample mean (Naive) or by the same calibrated model (Random:mg), and the break-even point against Naive. Uniform subsampling needs no panel, so it is the fairer comparison: EOST-M saves about 78 items per system against it and breaks even after a median of 122 new systems (from 67 to 352). When the calibrators' results already exist—public

leaderboards or earlier internal evaluations—the panel is free and adaptive testing is cheaper from the first system; when items are expensive, as in agentic benchmarks, the per-system saving dominates quickly. Calibrating a panel and running a 100-item test take seconds on one CPU core.

Table 6: Cost of adaptive evaluation with a fresh panel of eight calibrators.

Benchmark	I	break-even vs full evaluation	items to match Naive@100	break-even vs Naive	items to match Random:mg@100
GSM8K	1319	8.7	70	352	80
ARC-C	1172	8.7	20	117	90
HellaSwag	2000	8.4	8	174	25
WinoGrande	1267	8.7	20	127	60
MMLU	2000	8.4	15	188	100
SWE-bench V.	500	10.0	40	67	70
LiveCodeBench	1055	8.8	25	113	60
CRUXEval-O	800	9.1	25	85	60

10 Discussion

Small-panel calibration uncertainty is the problem. With four or eight calibrators a large fraction of items is answered identically by every calibrator, so their difficulty is located only by the prior. Plug-in testing treats these poorly located parameters as known: Fisher selection credits such items with information they do not carry, the plug-in posterior becomes overconfident, and the predicted outcomes of the unobserved items inherit the parameter errors—the bias floor of Figure 3. The expected response function caps the information of such items (Proposition 5) and predicts their outcomes with the uncertainty they actually have. Both effects fade as the panel grows (Figure 2): for the very large panels of recent work (Li et al., 2026), plug-in Fisher information is an adequate approximation; for new, private or expensive benchmarks, self-generated item pools, or the panels of four to eight systems that a single laboratory can afford, it is not.

Integrating and constraining are two remedies. Integrating the item posterior (EOST-M) predicted better than constraining the discriminations to a common value (C1), more so with sixteen calibrators, whereas the constrained model's intervals were better calibrated because they are wider, not because its estimates are more accurate. A small panel must therefore not be trusted with item-specific discriminations: integrating their uncertainty is preferable when prediction and score accuracy matter most, and constraining them gives a simpler and more conservative test.

Why not always use a fully Bayesian adaptive test? It is the principled answer to calibration uncertainty, and Fink et al. (2025) found it the most accurate correction with hundreds of calibration responses per item. With eight calibrators it did not help measurably, although equivalence was not shown either. By Lemma 1, EOST-M is the fully Bayesian model with the calibrators' abilities and the hyperprior held fixed; in this regime what matters, as far as our data can tell, is the item-level uncertainty that both integrate. A fully Bayesian test also needs a Markov chain per panel, whose tuning is a

source of error in its own right: our frozen burn-in was too short on some banks, and even an exact sampler with collapsed moves did not meet our convergence criterion. EOST-M is the simpler default. Two leads remain open: the near-converged sampler suggests a possible advantage of Bayesian integration for estimating the score, and panel-level uncertainty may matter more when the calibrators' abilities are poorly determined, as with short banks.

The criterion matters less than first hypothesized. The pilot program and study 1 agree on the direction of the original thesis—EIG learns more about competence than predictive entropy—but bound its size: under the same calibration-marginal model the gain on held-out prediction is small, and the information criteria are practically interchangeable (a descriptive comparison; the failure of H1 is not a test of equivalence). Corollary 4 explains why: entropy and information disagree only through the spread of discriminations among items with similar success probabilities, which the expected response function shrinks, and with a common discrimination the difference vanishes. For the bank score, Proposition 7 shows that uncertainty sampling becomes the right one-step criterion once competence is pinned down relative to the unobserved bank, and the score-error curves show both regimes. What an adaptive test for LLMs must get right is the treatment of calibration uncertainty, not the choice among information criteria.

Honest uncertainty. None of the four remedies tried in study 2—integrating the calibrators' abilities, leave-one-calibrator-out recalibration, mixing random items into the test, recalibrating a constrained model—gave nominal coverage with intervals as narrow as EOST-M's. The coverage decomposition and the prior predictive checks locate the problem where adaptive selection meets a model that is misspecified in the dependence among real systems' verdicts. Recalibration is not yet a validated remedy: it raised coverage, but neither frozen criterion for it was met, and on the well-specified synthetic bank it moved coverage away from the nominal level. Practitioners should report model-based intervals with the panel size and their expected under-coverage, and use a Wilson interval from uniformly sampled items when nominal coverage matters more than width.

Adaptive testing or benchmark reduction? With a handful of calibrators, tinyBenchmarks, ATLAS-style CAT and MINCE rely on quantities that a small panel cannot determine well. In this regime our evidence favours adaptive selection under calibration uncertainty, in the same direction on every benchmark but conclusively by the frozen criterion only against MINCE; with large panels the differences should shrink.

Limitations. (i) All evidence replays recorded verdicts: replay is exact for the recorded data but omits the sampling variability of stochastic decoding, and the protocol we release for live evaluation could not be run in our environment; the temporal validation approximates, but does not replace, a prospective deployment. (ii) LiveCodeBench and CRUXEval-O carry binarized pass rates; graded responses would need another measurement model. (iii) The unidimensional 2PL is misspecified in the dependence among real systems' verdicts. (iv) Strict clusters are a heuristic that may miss distant relatives such as distilled models, although R2 and C1 did not depend on how they were built. (v) Recalibration rests on very few pseudo-targets, without a finite-sample guarantee. (vi) A fully converged Bayesian comparator, for instance by Hamiltonian Monte Carlo, remains to be run. (vii) The analysis plans were frozen by hashing and committing them, without an external time stamp. (viii)

Item exposure was not controlled; an adaptive benchmark that serves the same informative items to every model would accelerate their contamination (Wainer, 2000). (ix) Joint MAP calibration made the planned one-parameter sensitivity analysis degenerate; marginal maximum likelihood would avoid this. (x) Self-generated items were studied only in a historical pilot. (xi) The cross-benchmark meta-analysis is secondary and limited by shared systems.

Recommendations. With a small calibration panel: (1) do not plug in item-specific discriminations calibrated on fewer than a few dozen systems, but integrate their uncertainty with an empirical-Bayes prior, or constrain them when a simple, conservative test matters more than accuracy; (2) select by any information criterion computed from the marginal model, and let the score-targeted criterion take over late in the test when the estimand is the score; (3) prefer the calibration-marginal model to the Bayesian treatments of the calibrators that we ran, which made no significant difference; (4) report model-based intervals with the panel size and their expected under-coverage; (5) exclude the target's whole cluster from its panel and analyse with clusters as units; (6) count the panel in the cost.

11 Conclusion

Adaptive testing can make LLM evaluation cheaper, but with the small calibration panels of new, private or expensive benchmarks it works only if the uncertainty of the item calibration is taken seriously. Integrating it through expected response functions with an empirical-Bayes prior is simple, exact given the calibrators, and caps the information of items that a small panel cannot locate. Under frozen analysis plans, this predicted held-out verdicts better than plug-in Fisher-information testing and a regularized Rasch-type test, estimated scores more accurately than uniform subsampling and MINCE, was ahead of random items, ATLAS-style testing and tinyBenchmarks on every benchmark without meeting the frozen criterion for these comparisons, and was not significantly different from Bayesian CATs that also integrate the calibrators' abilities, without being shown equivalent. The thesis that information rather than entropy should be maximized holds in sign but is modest once uncertainty is integrated. The open problem is honest uncertainty: model-based intervals remain too narrow, and recalibration from the panel improved coverage only at the price of width.

Data and code availability

The analysis code, the Docker environment, both frozen analysis plans, the scripts that rebuild every response matrix from its pinned public source, all experimental artifacts with their SHA-256 hashes, and the manuscript sources, including an extended version of this article with every secondary table and figure, are publicly available at `https://github.com/mesellemahmed/eost`. Response matrices derived from MIT-licensed sources are distributed with the code; the others are rebuilt from their pinned sources by the provided scripts and checked against the published hashes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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